Laboratory 3
Modeling and Design of PID Controllers

Preparation:

- Read the instructions carefully and make sure you are prepared (for example, theory wise) to complete the lab. You should solve all exercises indicated with 'Preparation'.
- We recommend that you revise the following concepts: linearization, transfer function, characteristic polynomial and pole placement.
- Download the file `laboration3.zip` from the course's Canvas room, unzip and save the files in your work folder.
- Open the model `laboration3.slx` in MATLAB and verify that it looks like the one in Figure 8.
- Take with you the optimal values of K, T_i and T_d obtained in Laboratory 2.

1 Introduction

In the previous lab, you had plenty of opportunities to investigate how the different parameters — also called tuning parameters — of a PID-controller affect the closed-loop performance of the system. You might have noticed that it can be challenging and tedious to select K, T_i and T_d by trial and error. In this laboratory, you will use systematic methods (placing the poles, dominant poles, etc.) from the automatic control theory to tune PID-controllers. In order to do this, first you need a mathematical model, i.e. a transfer function description of the closed-loop system, and then determine the control design criteria.

Laboratory equipment

The same process as in Laboratory 2 is used. If possible, use exactly the same apparatus and setup that you used before (Figure 1). Since the pump and tube properties might be slightly different, it might be more difficult to compare your results to previous results if a different setup is used.

2 Modeling

The first step is to model (define how states vary over time) the system. Remember the simplified train model you implemented in Simulink in Laboratory 1. That model, comprised of two differential equations, was obtained by equating forces acting on the train masses and sketching a free-body diagram. Here, we are also interested in calculating differential equations that govern how variables of interest (or states) vary over time in the tank process. Observe that the actual tank process you use in the experiment comprises of two tanks connected in series. In this section, we focus on modeling a process with a single tank, but the results and conclusions are straightforward extended to the double tank case. It is also worth highlighting the difference between inputs and outputs in the two cases. For a single tank, the input is the water flow rate coming from the pump and the output is the tank level. For the double tank process, the input is also the water flowrate coming from the pump, and the output is also the level of the upper (or lower) tank. In the latter though, there are two 'internal models', one for each tank. The input of the upper tank is the water flowrate coming from the pump and the output in the upper tank level. The input of the lower tank is the water flowrate coming from the **upper tank**, and the output is the level of the lower tank.

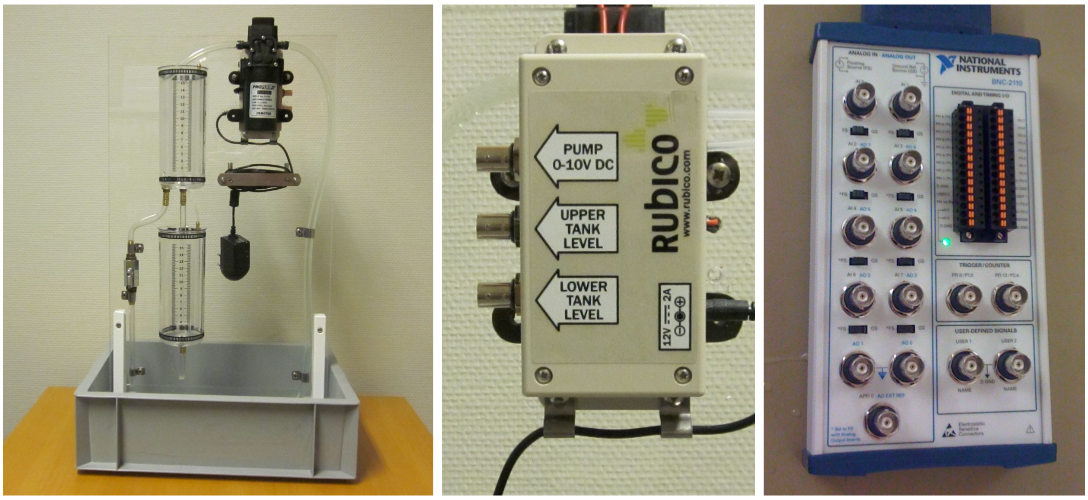


Figure 1: Double tank process and connector blocks.

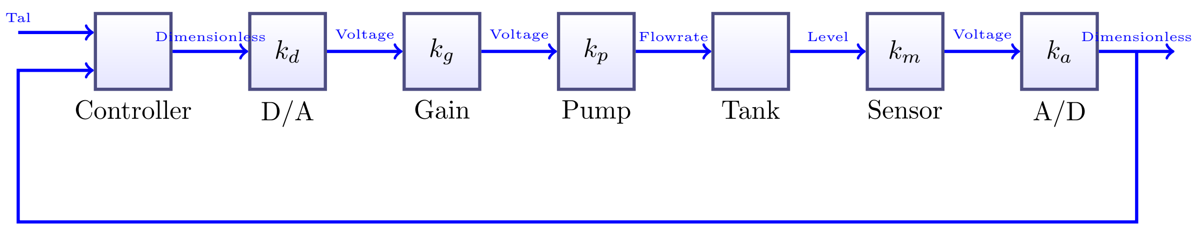


Figure 2: Block diagram of the double tank closed-loop system.

Start by identifying all the relevant components of the closed-loop system and sketching a block diagram as in Figure 2. The discrete control signal generated by the computer is converted to a voltage signal (D/A, digital to analog) and sent to the pump. The pump responds by generating a water flowrate. The sensor reads the water level in the tank and sends a voltage signal back to the computer, which is converted first to a digital signal (A/D, analog to digital).

Very often the signals in different units in Figure 2 can cause confusion and thus lead to errors during the controller design stage that could be easily avoided by further simplifying the block diagram in Figure 2 to the one in Figure 3. The main difference between the two diagrams is that some blocks have been lumped together to simplify unit conversions. Therefore, to implement the double tank model we need to calculate the values of α and β and define the relationship between incoming water flowrate and tank level.

Preparation 2.1 *What are the dimensions of $\alpha = k_d k_g k_p$ and $\beta = k_m k_a$?*

Model of a single tank system

The equivalent of free body diagram (used for the train model in Laboratory 1) for fluid systems are the continuity (or transport) equations. For a single tank, the mass in the tank (or volume assuming

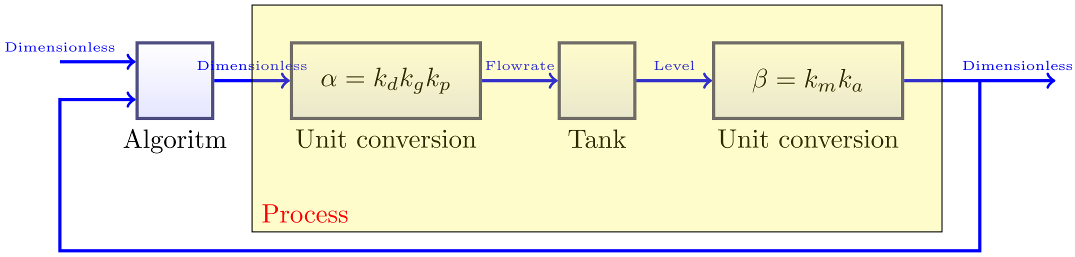


Figure 3: Block diagram of the closed-loop system.

constant density) varies over time according to:

$$Change\_in\_volume = inflow - outflow \tag{1}$$

The water inflow comes directly from the pump and can be treated as the external signal $q_u(t)$ [m^3/s]. The tank level is $h(t)$ [m]. According to Torricelli's law, the relationship between the water velocity at the outflow ($v(t)$ [m/s]) and the water level inside the tank is given by

$$v(t) = \sqrt{2gh(t)} \tag{2}$$

where $g = 9.81$ m/s^2 is the acceleration due to gravity. Assuming that the tank has a constant cross-section area A [m^2] and that the cross-section area of the outflow tube (also constant) is a [m^2], the water outflow $q_i(t)$ [m^3/s] is calculated as:

$$q_i(t) = av_i(t) \tag{3}$$

Preparation 2.2 *Use the Torricelli's law and mass balance to write two differential equations that define the relationships between:*

1. *water inflow from the pump $q_u(t)$ and water level in the upper tank $h_1(t)$.*
2. *water level in the upper tank $h_1(t)$ and water level in the lower tank $h_2(t)$. Observe that the water flowrate through the outflow tube from the upper tank is proportional to the water flowrate velocity through the tube (see equations above).*

Now you should have a tank model with flowrate [m^3/s] as input and tank level [m] as output (and state). Before starting to design a controller, we need to further simplify this model so that the inputs and outputs are ideally dimensionless and normalized in the interval $(0, 1)$. Normalization is not mandatory, but it makes calculations easier in later stages. It is important that you understand the difference between state and output. In short, outputs are measured states.

Preparation 2.3 *Calculate a dimensionless tank model by implementing the following simplifications: $q_u(t) = \alpha u(t)$, $h_1(t) = x_1(t)/\beta$ and $h_2(t) = x_2(t)/\beta$ with parameters from Preparation 2.1. The resulting states of the model should be $\{x_1, x_2\}$, equivalent to tank levels $\{h_1, h_2\}$, normalized in the interval $(0, 1)$. Observe that the equations in the original (with dimensions) model will be multiplied by appropriate parameters that make the new model dimensionless. The next Preparation Exercise helps develop the normalized model.*

Preparation 2.4 *Demonstrate that the normalized double tank model is given by:*

$$\begin{aligned} \frac{dx_1(t)}{dt} &= -\gamma_1\sqrt{x_1(t)} + \delta u(t) \\ \frac{dx_2(t)}{dt} &= \gamma_1\sqrt{x_1(t)} - \gamma_2\sqrt{x_2(t)} \end{aligned} \tag{4}$$

where

$$\gamma_1 = \frac{a_1}{A}\sqrt{2g\beta} \quad [\mathrm{s}^{-1}]$$
$$\gamma_2 = \frac{a_2}{A}\sqrt{2g\beta} \quad [\mathrm{s}^{-1}]$$
$$\delta = \frac{\alpha\beta}{A} \quad [\mathrm{s}^{-1}]$$

The above model is a nonlinear state-space model (can you guess why?). The interval $(0, 1)$ in the model states x_1, x_2 is equivalent to the interval $(0, 0.15)$ [m] in the upper and lower tank levels.

The theoretical values of δ, γ_1 and γ_2 can be calculated using supplier data. This includes for example values of A and a.

Cross-section area, tank:	$A_i = 3.5 \times 10^{-3}\,\mathrm{m}^2$
Cross-section area, tube:	$a_i = 1.96 \times 10^{-5}\,\mathrm{m}^2$
Control signal to flowrate:	$\alpha = 4.9 \times 10^{-5}\,\mathrm{m}^3/\mathrm{s}$
Tank level to measured variable:	$\beta = 6.67\,\mathrm{m}^{-1}$

Preparation 2.5 *These theoretical values above are usually only accurate if the supplier data is accurate, which might not be the case given for example wear and tear over time. On the other hand, it is relatively easy to measure these values with simple experiments in this laboratory. Observe that in practice, measurement error can also lead to poor estimation of model parameters, and it might be impossible to know what is the 'right answer', since the theoretical values might be outdated.*

The following experiments can be used to calculate the values of the model parameters in practice:

- *Estimate δ by setting a constant pump flowrate, fully blocking the outflow tube in the upper tank (make sure that the disturbance valve is closed) and measuring the time it takes to fill the tank from one level to another. Observe that the level sensor does not register values in the interval $(0, 0.1667)$, so start from a tank level higher than 0.2.*
- *Estimate γ_1 and γ_2 by setting a constant water inflow, wait until the process is in steady-state, and measure the equilibrium states x_1^0 and x_2^0.*

Demonstrate how the above experiments (and their collected data) and Eq. (4) can be used to actually estimate the model parameters.

Show how, using the above experiment and Eq. (4), one can calculate the experimental value of δ, γ_1 and γ_2, in the corresponding order.

Exercise 2.1 *Do the experiments and calculations explained in Preparation 2.5 to calculate the experimental values of δ, γ_1 and γ_2. Compare the theoretical and experimental values. They should be somewhat close to each other. Tip: the Simulink model* `lab2timer` *might help you in this experiment.*

	Theoretical value	Experimental value
δ		
γ_1		
γ_2		

Now you should have a normalized model with dimensionless inputs, outputs and states. All the required model parameters have been calculated experimentally and we don't have to worry about dimensions anymore.

3 Linear model

The model in Eq. (4) in a nonlinear state-space model. Although there are (more advanced) methods to tune PIDs using non-linear models, we are going to do it the other way around: use a more simple tuning method that only works for linear models. Therefore, our next step is to linearize the model in Eq. (4).

Preparation 3.1 *Linearize the tank model* (4) *around a general operating point* (x_1^0, x_2^0).

Preparation 3.2 *Demonstrate that the linearized model from the Preparation 3.1 is equivalent to the following transfer functions:*

$$\begin{aligned}\Delta Y_1(s) &= \frac{\rho\tau_1}{1+s\tau_1}\Delta U(s)\\ \Delta Y_2(s) &= \frac{\rho\tau_2}{(1+s\tau_1)(1+s\tau_2)}\Delta U(s)\end{aligned} \tag{5}$$

Determine the parameters ρ*,* τ_1 *and* τ_2 *as a function of the model parameters* δ*,* γ_1 *and* γ_2 *at the operating point* (x_1^0, x_2^0).

4 Designing PID-controllers by placing the closed-loop poles

Now that we have at our disposal, a linearized model of the process we want to control, we can proceed to use a systematic method to design the controller(s). Observe that one might wonder what is easier: to just run trial and error experiments to find acceptable PID parameters (as in Laboratory 2), or to follow the methodology proposed in this lab. In practice, it is not uncommon that operators use a combination of default settings and trial and error to tune industrial PIDs. Nevertheless, as a student you need to learn the method described here to pass the course, but we (teachers) hope that by the end of the laboratory, you can reflect on pros and cons of the two approaches and be able to suggest one or the other based on the goal and circumstances.

We will design two PID controllers, one for the upper tank and another for the double tank process. Observe that these two different processes have two different models. The first step of the design method is to define the desired properties of the closed-loop response. Previously (in Laboratory 2), we used qualitative descriptions of performance such as fast response, not too aggressive, small overshoot, no offset. The design method allows for much more precise description of performance in terms of the poles of the closed-loop transfer function, which are directly related to time domain properties such as rise time, settling time and maximum overshoot.

Consider a second-order transfer function

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2}$$

and a unitary step input signal ($U(s) = 1/s$)). The output in the time domain is

$$y(t) = 1 - \frac{e^{-\zeta\omega\ t}}{\sqrt{1-\zeta^2}}\ \sin\left(\omega\sqrt{1-\zeta^2}\ t + \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right).$$

The choice of ζ and ω in the Laplace domain (position of the closed-loop poles) defines the properties of the output response in the time domain.

Preparation 4.1 *Consider a process represented by the following under-damped transfer function:*

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2},\ \ 0 < \zeta < 1.$$

Write the process's poles as a function of ζ and ω.

Preparation 4.2 *Calculate the process' rise time (t_d), settling time ($t_{5\%}$), maximum overshoot (M) and time for maximum overshoot (t_M) as a function of the parameters ζ and ω for $U(s) = 1/s$ (unitary step input).*

Observe that in the above Preparation Exercises, the example transfer function is a generic transfer function $G(s) = Y(s)/U(s)$. In the Modeling Section of this laboratory, you defined process transfer functions for the upper tank (process) and the upper + lower tank (process). In the open loop configuration, The controller has its own transfer function, generically described as $G_c = U(s)/R(s)$, where $U(s)$ is the control signal and $R(s)$ is the reference signal $R(s)$. However, the controller we use here require a closed-loop configuration. Soon a new transfer function — the closed-loop transfer function, a 'combination' of process and controller — will be introduced. The transfer function of the PID controller in the closed-loop system is generically described as $G_c = U(s)/E(s)$, where $U(s)$ is the control signal and $E(s)$ is the error signal, which is the difference between the reference signal $R(s)$ and the output signal $Y(s)$.

Now we have an idea about how the pole locations of a transfer function correlate to performance in the time domain. We can start designing PI and PID controllers using the pole placement method.

The method is based on the assumption that if by choosing the parameters in the controller in a suitable way, one can make the closed-loop characteristic polynomial (denominator of the closed-loop transfer function) as seen in Figure 4, look exactly the same as a design (or desired) characteristic polynomial, then the time domain performance of the closed-loop system will (almost) be the same as the time domain behavior of the design polynomial. Observe that this is an indirect way of calculating the PID parameters (K, T_i and T_d) by defining the location of the poles in the desired polynomial (in the case of a second order polynomial, ω and ζ).

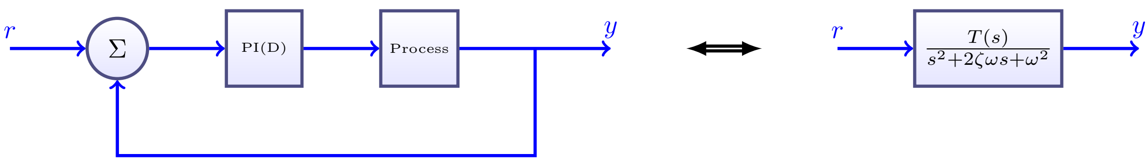


Figure 4: In the pole placement method, the closed-loop system's properties are defined by specifying the properties of a desired transfer function (and thus characteristic polynomial). $T(s)$ in a transfer function that depends on the process and controller.

Automatic control of the upper tank

Preparation 4.3 *Use the equation of the upper tank from the process model in Eq.* (5) *from Preparation Exercise 3.2 to design a PI-controller of the form*

$$u(t) = K\left(e(t) + \frac{1}{T_i}\int_0^t e(\tau)d\tau\right) \quad \Leftrightarrow \quad U(s) = K\left(1 + \frac{1}{sT_i}\right)E(s) \tag{6}$$

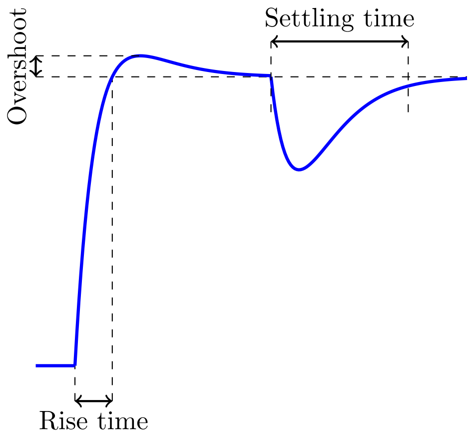


Figure 5: Definition of rise time and overshoot for setpoint tracking and settling time for disturbance rejection.

and control the upper tank. Calculate appropriate tuning parameters for the PI-controller (K and T_i) so that the closed-loop system has a relative dampening ζ and natural frequency ω, and the following closed-loop characteristic polynomial:

$$s^2 + 2\zeta\omega s + \omega^2 \tag{7}$$

In your answer, the PI parameters K and T_i should be written as a function of the process parameters ρ and τ_i and the controller design parameters ω and ζ.

Tips: *For a open-loop transfer function (controller × process) $G_0 = Q/P$ where P and Q are (relatively prime) polynomials, the closed-loop transfer function can be written as:*

$$G = \frac{G_0}{1+G_0} = \frac{Q/P}{1+Q/P} = \frac{Q}{P+Q} \tag{8}$$

The characteristic polynomial is $P + Q$.

Exercise 4.1 *First, calculate the PI parameters for the upper tank with $\zeta = 1$ and $\omega = 0.3$ and operating point $x_1^0 = 0.5$. You can use the MATLAB script* `picalc` *as follows:*

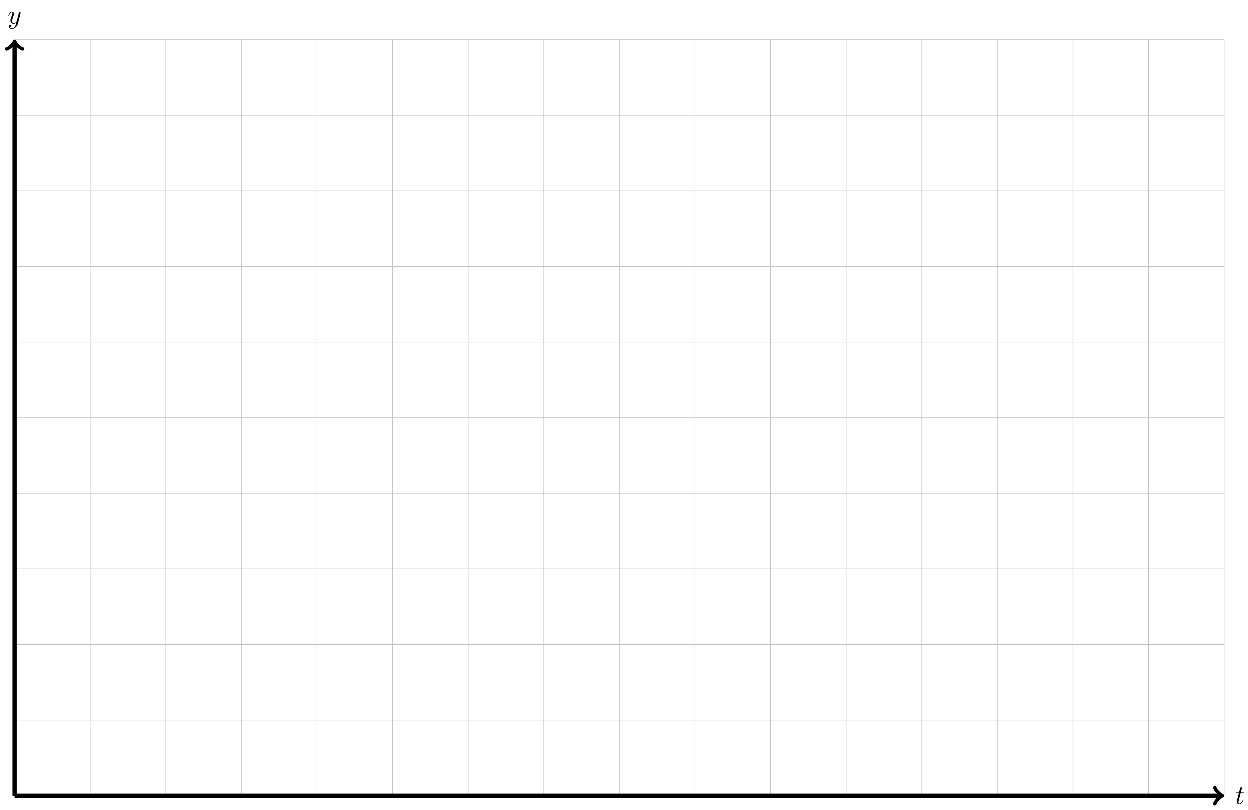
```
>> delta = ...;
>> gamma1 = ...;
>> gamma2 = ...;
>> zeta = 1;
>> omega = 0.3;
>> picalc
K = 5.906    Ti = 6.169
```

Compare your calculations (from the Preparation Exercise) with ones implemented in picalc. (`type picalc`).

Exercise 4.2 *Now we need to analyze the actual performance of the designed PI-controller with respect to setpoint changes and disturbance rejection. Do the following experiment:*

1. *Set a PI controller to control the level in the upper tank in the Simulink model.*
2. *Enter the calculated PI parameters K and Ti.*
3. *Enter the setpoint $r = 0.5$ and wait until the tank system is stable.*

4. *Change the setpoint to $r = 0.7$ and sketch the output response in the Figure below. Plot both the control signal and the water level from your Simulink experiment.*
5. *Return the setpoint to $r = 0.5$ and wait until the system is stable.*
6. *Open the disturbance valve 50% and sketch the output response in the Figure below (what happens if you open the disturbance valve completely?). Show both the control signal and the water level from your Simulink experiment.*



Tips: *Your curves should resemble the ones in Figure 5. If you are unsure about your results, discuss them with one of the lab assistants.*

Poles and zeros

The block diagram in Figure 6 represents the upper tank in closed-loop. The closed-loop transfer functions between the reference $R(s)$ and the output $Y(s)$ and the disturbance $L(s)$ and the output $Y(s)$ are given by:

$$G_{YR}(s) = \frac{G_p(s)G_r(s)}{1 + G_p(s)G_r(s)} = \frac{\rho K(s + \frac{1}{T_i})}{s^2 + s(\frac{1}{\tau_1} + \rho K) + \frac{\rho K}{T_i}} \tag{9}$$

$$G_{YL}(s) = \frac{G_p(s)}{1 + G_p(s)G_r(s)} = \frac{s\rho}{s^2 + s(\frac{1}{\tau_1} + \rho K) + \frac{\rho K}{T_i}} \tag{10}$$

By using the pole placement method, and defining the desired polynomial as a second-order transfer function, we can properly define both K and T_i by comparing the denominator of the closed-loop transfer function and the desired polynomial. Observe that the nominator of the closed-loop transfer function are also a function of K and T_i.

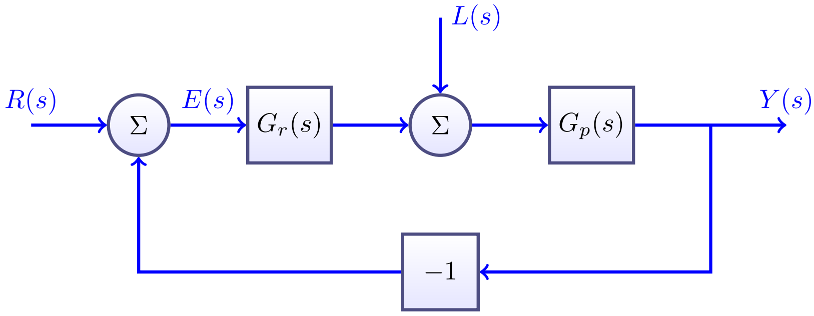


Figure 6: Block diagram representing the closed-loop control of the upper tank.

By analyzing $G_{YR}(s)$ in Eq. (9), we can see that one zero (root of the nominator) in this transfer function is a function of T_i. However, such dependency is **not** found in $G_{YL}(s)$. This is the reason why overshoot is sometimes observed in the setpoint tracking response, even though the design polynomial has $\zeta = 1$ (critically damped response, and thus no overshoot).

The takeaway is that in order to investigate and validate controller performance after tuning, one should firstly consider the disturbance rejection response if only poles are important, but also consider the setpoint tracking response if both poles and zeros are important.

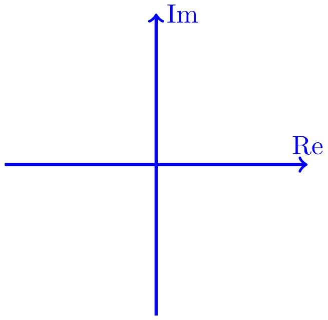
Exercise 4.3 *Now we can systematically analyze how the closed-loop pole locations (values of ζ and ω) affect the closed-loop performance by comparing different parameter values..*

Start by setting $\zeta = 1$ and vary ω according to the Table below. Use $x_1^0 = 0.5$ as the initial operating point. Calculate the values of K and T_i for different values of ω, complete the experiments according to the instructions in Exercise 4.2 and complete the Table, root locus (location of the poles of the closed-loop system) and Figures below.

In the end, try the optimal PI parameters from Laboratory 2. You can calculate backwards the values of $\zeta = 1$ and ω based on K and T_i by simply reversing the formulas, if you are curious. Perform the experiment again and report the results in the last row of the Table 1. Plot both the control signal and the water level from your Simulink experiment.

Table 1: Experiment summary, Exercise 4.3.

				Setpoint change			Disturbance
ω	ζ	K	T_i	Rise time [s]	Overshoot [%]	Saturation in the control signal (yes/no)	Settling time [s]
0.1	1						
0.2	1						
0.5	1						
Lab 2	Lab 2						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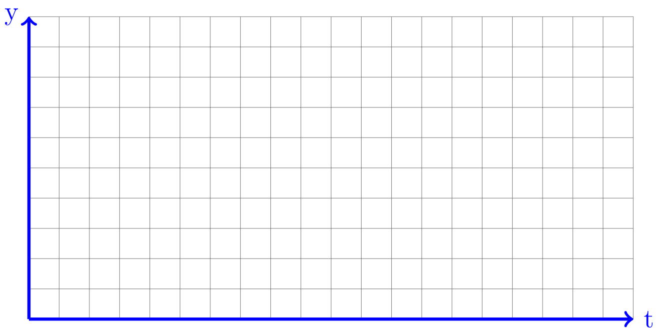
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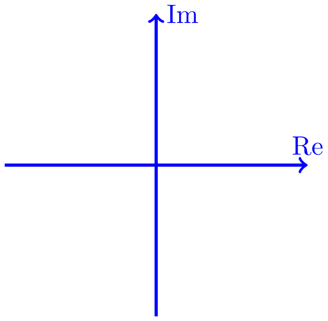
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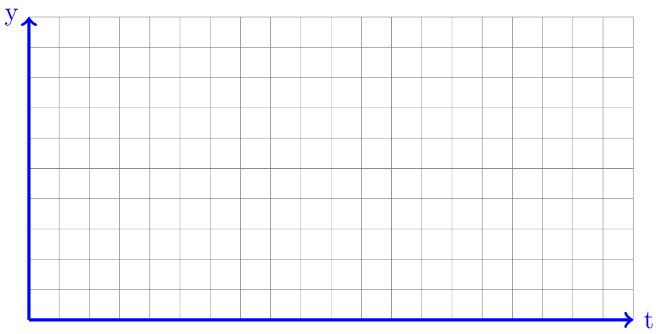
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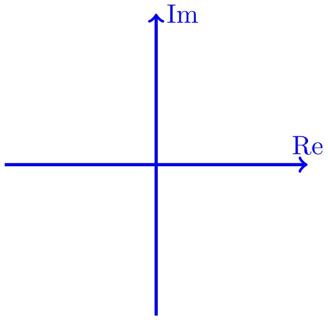
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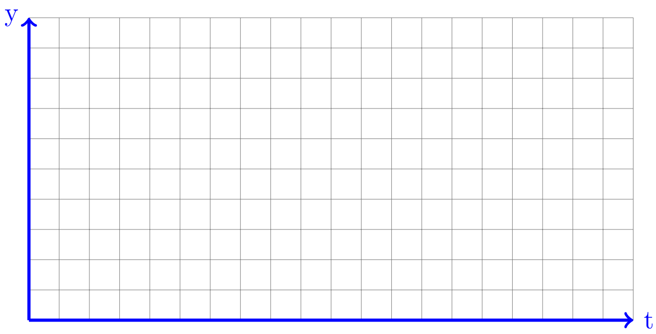
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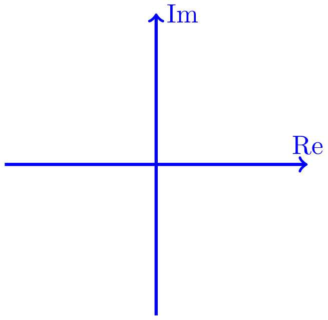
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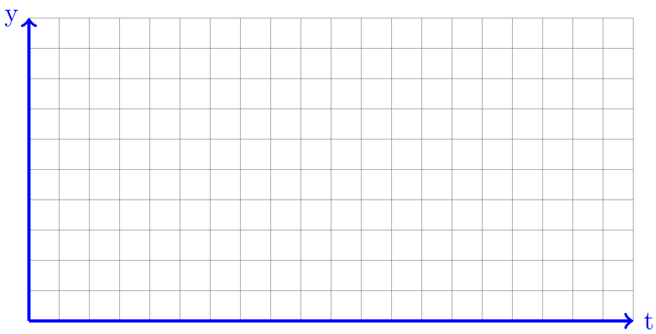
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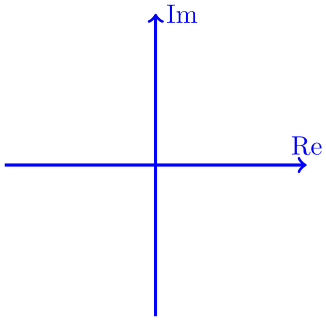
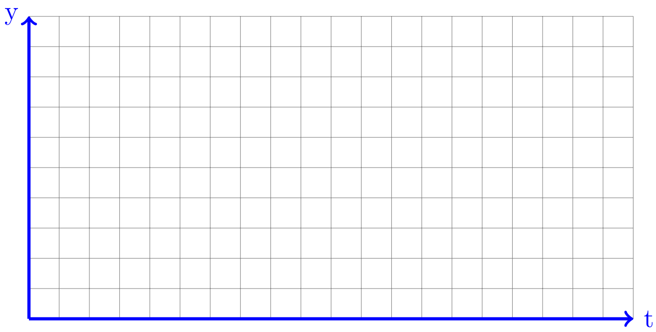
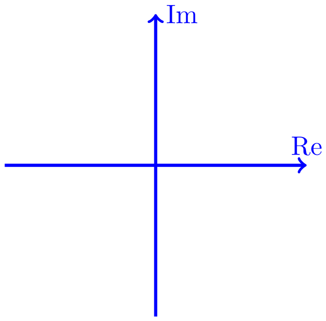
Exercise 4.4 *Repeat the experiment from Exercise 4.3 but use a constant* $\omega = 0.4$ *and vary* ζ *according to the Table below. Fill in the Table, Root Locus and time responses. Observe how the response in the time domain changes with different values of* ζ*, in particular the oscillation. Plot both the control signal and the water level from your Simulink experiment.*

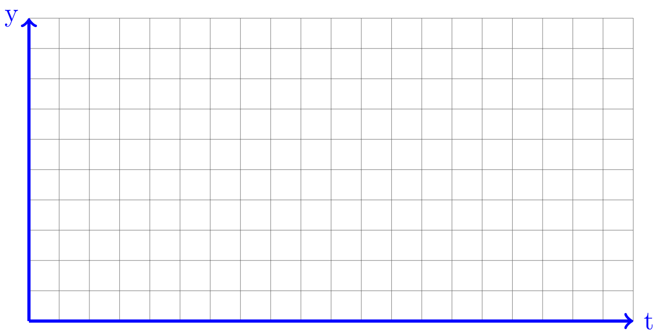
Table 2: Experiment summary, Exercise 4.4.

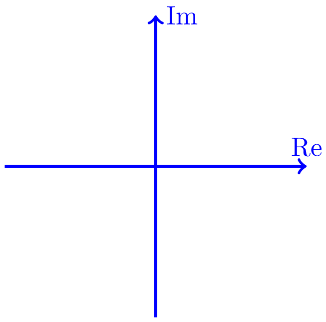
				Setpoint change			Disturbance
ω	ζ	K	T_i	Rise time [s]	Overshoot [%]	Saturation in the control signal (Yes/No)	Settling time [s]
0.4	0.7						
0.4	0.4						
0.4	0.2						

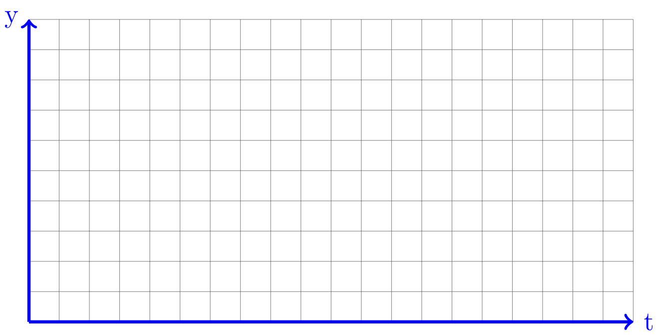












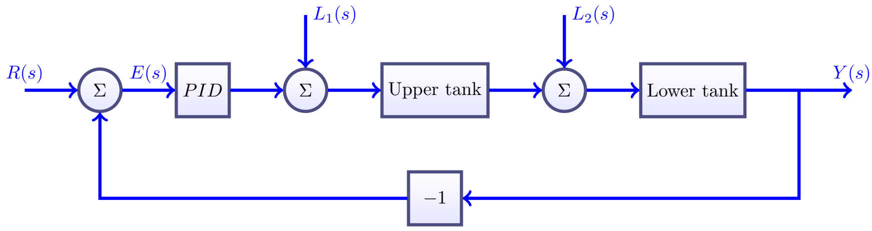


Figure 7: Block diagram representing the closed-loop control of the lower tank.

Automatic control of the lower tank

Preparation 4.4 *Use the second transfer function in Eq.* (5) *to design a PID-controller*

$$u(t) = K\left(e(t) + \frac{1}{T_i}\int_0^t e(\tau)d\tau + T_d\frac{de(t)}{dt}\right) \quad \Leftrightarrow \quad U(s) = K\left(1 + \frac{1}{sT_i} + sT_d\right)E(s) \tag{11}$$

to control the tank level in the lower tank. Observe that previously, the single tank process has one pole and the controller (PI) also has one pole, resulting in a second-order (two poles) characteristic polynomial. In this case, the process (double tank) has two poles and the controller (PID) has one pole, resulting in a third-order (three poles) characteristic polynomial. Therefore, even the design polynomial has to be of third order. It is not straightforward to derive a relationship between properties in the time domain and poles of a third-order transfer function, so to simplify the design method, we will take advantage of the concept of dominant poles. Dominant poles are defined as the poles that contribute the most to the transient behaviour of the closed-loop system. A simple way to visualize this is to compare the effect of stable poles with respect to how close they are to the origin. Comparing the poles $p_1 = -0.5$ *and* $p_2 = -5$, *we can say that the pole closest to the origin (*p_1 *in this case) is dominant since the transient response of* p_2 *'is over' much quicker than that of* p_2. *Another way to understand this is to go back to the definition of time constant in first-order systems. For a first-order system* $G(s) = 1/(\tau s + 1)$, τ *is defined as the time the output takes to reach 63% of its steady-state value for a step input. The pole of* $G(s)$ *is* $p = -1/\tau$. *Therefore, is* τ *is large, then the pole is close to the origin and the output response is slow.*

Going back to the characteristic polynomial of the double tank system, we are going to choose complex conjugate dominant poles (second-order transfer function) and multiply it by a third pole that is much faster (far away from the origin) than the dominant poles.

The desired characteristic polynomial is:

$$(s + \alpha)(s^2 + 2\zeta\omega s + \omega^2) \tag{12}$$

Derive K, T_i *and* T_d *as a function of the model parameters* ρ, τ_1 *and* τ_2 *and the design parameters* ω, ζ *and* α.

Poles and zeros

The goal now is to design a controller for the lower tank (and thus for the combined upper and lower tank system). Figure 7 illustrates the block diagram of the closed-loop system. Disturbances indicated as L_1 and L_2 can affect the system in two different points: the upper tank (example filling in the upper tank with a cup of water or opening the disturbance valve) and the lower tank (example filling in the lower tank with a cup of water). The transfer functions between the reference and output (G_{YR}) and disturbances and output (G_{YL_1}, G_{YL_2}) are:

$$G_{YR} = \frac{K\rho(s^2\frac{T_d}{\tau_1} + s\frac{1}{\tau_1} + \frac{1}{\tau_1 T_i})}{s^3 + (\frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{K\rho T_d}{\tau_1})s^2 + (\frac{1}{\tau_1\tau_2} + \frac{K\rho}{\tau_1})s + \frac{K\rho}{\tau_1 T_i}} \tag{13}$$

$$G_{YL_1} = \frac{\frac{\rho}{\tau_1}s}{s^3 + (\frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{K\rho T_d}{\tau_1})s^2 + (\frac{1}{\tau_1\tau_2} + \frac{K\rho}{\tau_1})s + \frac{K\rho}{\tau_1 T_i}} \tag{14}$$

$$G_{YL_2} = \frac{s\frac{1}{\tau_1}(s + \frac{1}{\tau_1})}{s^3 + (\frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{K\rho T_d}{\tau_1})s^2 + (\frac{1}{\tau_1\tau_2} + \frac{K\rho}{\tau_1})s + \frac{K\rho}{\tau_1 T_i}} \tag{15}$$

Observe that the denominator of the transfer functions above is identical. Only the denominator of G_{YR} is affected by the choice of PID-parameters.

Exercise 4.5 *Now it is time to investigate how the closed-loop performance varies with different design criteria. Remember to activate the Integral and Derivative components of the controller and to set the measured output to the level in the lower tank before start experimenting.*

Set the dampening constant to $\zeta = 0.7$, the real (non-dominant) pole $-\alpha$ with $\alpha = 1$ and vary ω according to Table 3.

Note that the value of α is chosen freely, but it has to be such that $-\alpha$ is at least 5 times of the real value of the other poles (dominant poles), but not too large that may require too large control signal (which in this experiment, as the voltage to the pump is limited, will cause saturation to the control signal). If $\alpha = 1$ causes saturation to the control signal, which means the non-dominant pole is too far to the left, shift it closer to the imaginary axis. Find the values of the open-loop poles of the plant G_p (in this case the lower tank), and replace $-\alpha$ with $(5 \times s_{min})$, where s_{min} is the smallest, or most negative, of the two open-loop poles of G_p. This will give the value of $0 < \alpha < 1$.

Assume that the system starts from the operating point $x_2^0 = 0.5$ and calculate the PID parameters using the MATLAB script `pidcalc` *as in the following example:*

```
>> omega = 0.15;
>> zeta = 0.7;
>> alpha = 1;
>> pidcalc
K = 12.346    Ti = 15.415   Td = 5.211
```

Analyze the closed-loop behavior to setpoint changes and disturbances in L_1. Sketch the responses in the Figures below, fill in the Root Locus plots and observe how the location of the poles is related to the rise time in the time domain response.

Follow these steps:

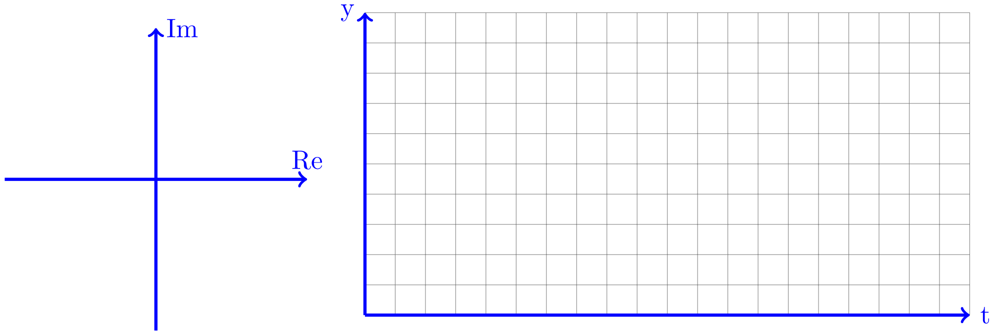
1. Activate I and D in the Simulink model
2. Make sure that the disturbance valve is closed.
3. Set the PID parameters K, T_i and T_d.
4. Set the reference to $r = 0.5$ and wait until the system is at steady state.
5. Change the setpoint to 0.7 and sketch the time response. Fill in Table 3 with overshoot, rise time and whether the control signal saturates. Plot both the control signal and the water level from your Simulink experiment.
6. When the system is at steady state again, open the disturbance valve at 50% and sketch the disturbance rejection response. Fill in Table 3 with the settling time. Plot both the control signal and the water level from your Simulink experiment.

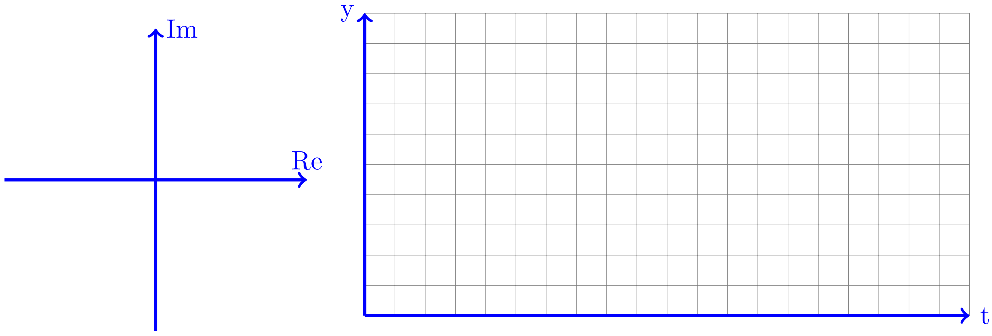
Table 3: Experiment summary, Exercise 4.5.

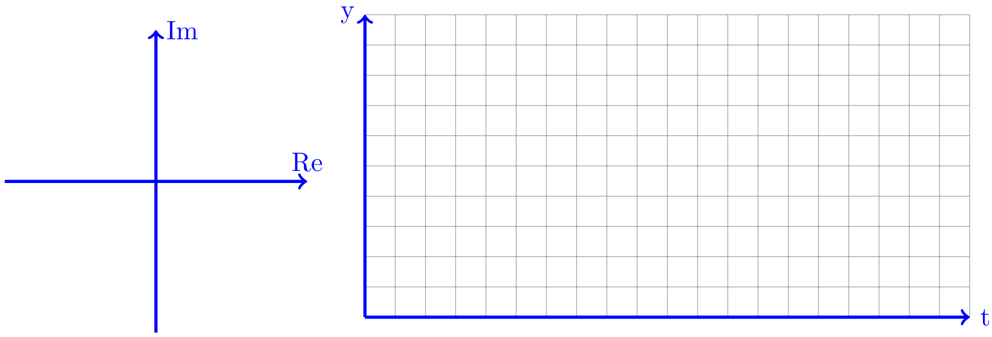
ω	ζ	K	T_i	T_d	Setpoint change			Disturbance
					Rise time [s]	Overshoot [%]	Saturation in the control signal (Yes/No)	Settling time [s]
0.1	0.7							
0.15	0.7							
0.2	0.7							
Lab 2	Lab 2							

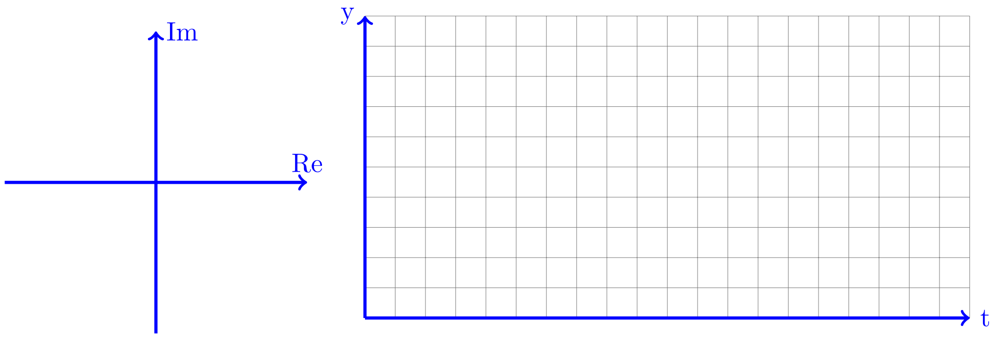
Finally, try your optimal PID parameter values from Laboratory 2 and fill in the last row of Table 3

Tips: This experiment can take time (waiting until the system is at steady state). You can start working with the questions in the Summary section to save time.





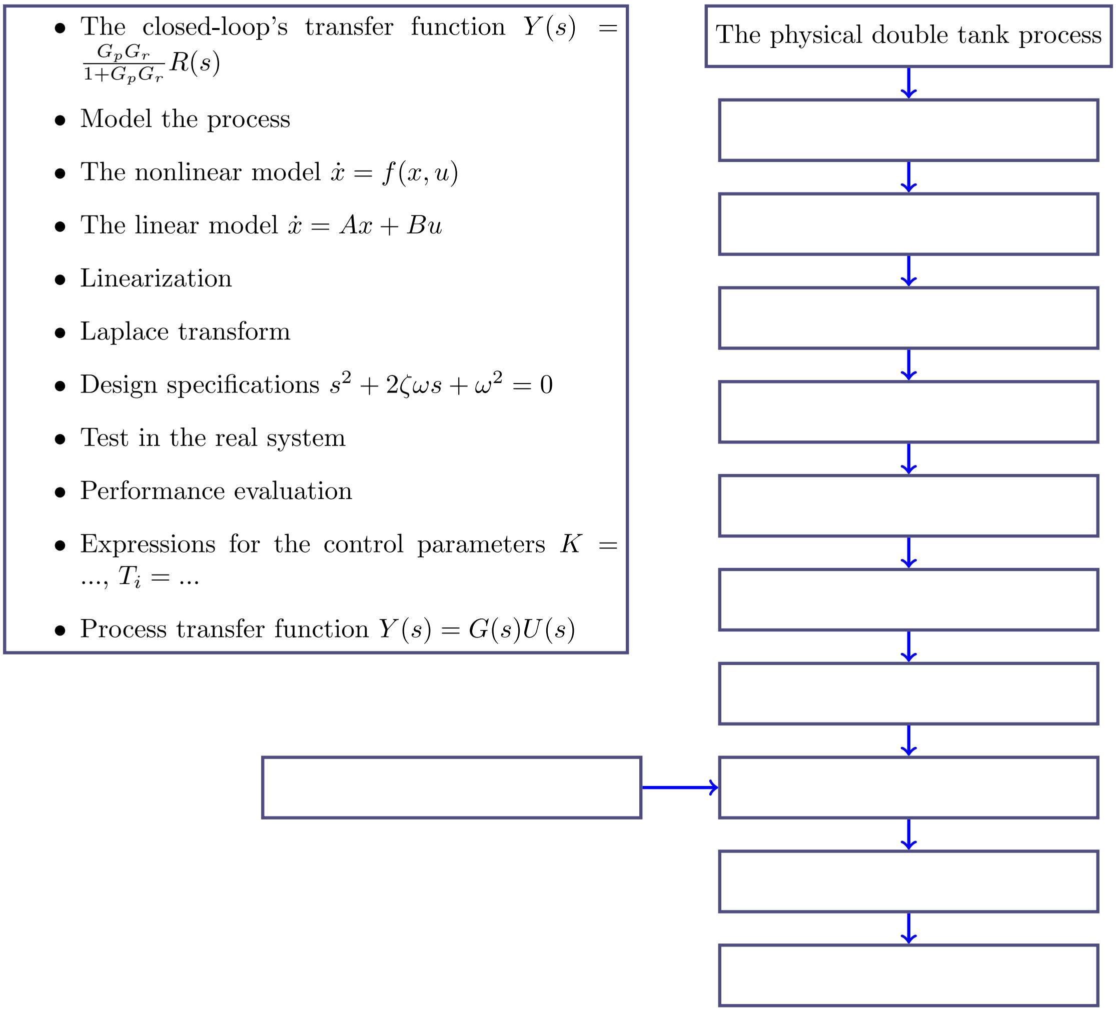




5 Summary

The purpose of this section is twofold: (i) to highlight one more time the usual workflow to design controllers using the pole placement method, and (ii) to present important questions that you should be able to answer after completing this Laboratory.

Exercise 5.1 *Complete the flowchart below with the activities done before and during the Laboratory. Observe that the step where you estimated the model parameters is not included. In what order should this step be completed?*



Exercise 5.2 *Write at least two characteristics of the real process that are not included in the process model.*

Exercise 5.3 *Considering the PI-controller in the upper tank, how does the parameter ω affect the poles of the closed-loop system (what happens if one increases/decreases ω)? What is the resulting effect in setpoint tracking and disturbance rejection?*

What happens with K and T_i when ω increases? Why don't we test $\omega = 10\,\mathrm{rad\,s^{-1}}$?

Exercise 5.4 *Considering the PI-controller in the upper tank, how does the parameter ζ affect the poles of the closed-loop system (what happens if one increases/decreases ζ)? What is the resulting effect in setpoint tracking and disturbance rejection? What would the closed-loop responses look like if $\zeta = 0$?*

Exercise 5.5 *Why the D-component is neglected during the controller design for the upper tank?*

Exercise 5.6 *Considering the PID-controller in the lower tank, how many poles does the closed-loop system have?*

Exercise 5.7 *Considering the PID-controller in the lower tank, how does the poles of the closed-loop system change when ω is increased? What is the resulting effect in setpoint tracking and disturbance rejection?*

Exercise 5.8 *Considering the PID-controller in the lower tank, how does the PID parameters K, T_i and T_d change when ω is increased? Why don't we test $\omega = 1\,\mathrm{rad\,s^{-1}}$?*

Exercise 5.9 *Fill in Table 5 with the optimal PID parameters obtained in Laboratories 2 and 3. Discuss pros and cons of the methods used to calculate them.*

	Upper tank	**Lower tank**
P	$K =$	$K =$
PI	$K =$ $T_i =$	$K =$ $T_i =$
PID	$K =$ $T_i =$ $T_d =$	$K =$ $T_i =$ $T_d =$

Table 4: Optimal PID parameters by pole placement and trial and error.

6 Evaluation

6.1 Oral session

The Oral session is an opportunity for you and your group to discuss the lab instructions with the lab assistants, and also an opportunity for us to make sure you have prepared to the lab. Check the Canvas room for information on how to book your session. The goal is not to evaluate your performance in the lab (which is done in a report), but rather to check that you know what you are going to do, how it is relevant to the course's theory and what is expected from you. In other words, this is a formative evaluation activity that should be more useful to you than to us.

6.2 Report

Write a report that briefly describes what you have done during the lab. Include figures that show how your final model in Simulink, as well as your phase portrait. Discuss your results, what you have learned and how you think they will be/are relevant to the course. The report grading is based on the following grading matrix:

Section	Points	
4		Report Format
	2	Neat and Organized
	2	Format of tables and figures is correct
4		Completeness of Report Components
5		Introduction & Preparation(aim, objectives, list of apparatuses, references)
12		Simulink Diagram Properness
	4	Correct diagrams
	4	Completeness of labelling
	4	Output display
10		Saving Data
	5	Correctness
	5	Choice of variables
20		Graph Plotting Elaboration
	5	Correctness of plotted data
	5	Completeness of properties
	5	Proportion of display (thickness, font size, etc.)
	5	Clarity
40		Results & Analysis
	10	Correctness
	10	Technical depth
	10	Quantitative analysis
	10	Presenting challenges, ideas for improvement, applications
5		Conclusion
	5	Relevance to the objectives
		Total Possible Points: 100

The report should be submitted in Canvas before the deadline.
Grade scale:
< 80 - U, ≥ 80 - G.

See the Canvas room for more information.

7 Review history

When	Who	What
2024-06-13	ASY	Instructions translated to English, minor improvements.
2024-03-22	ASY	New grading criteria.
2023-05-31	EKoJE	New Preparations Exercises to better understand how design polinomial and closed-loop polinomial are connected.
2023-03-06	ASY	New grading criteria.
2023-03-06	ASY	Minor improvements.

A Simulink

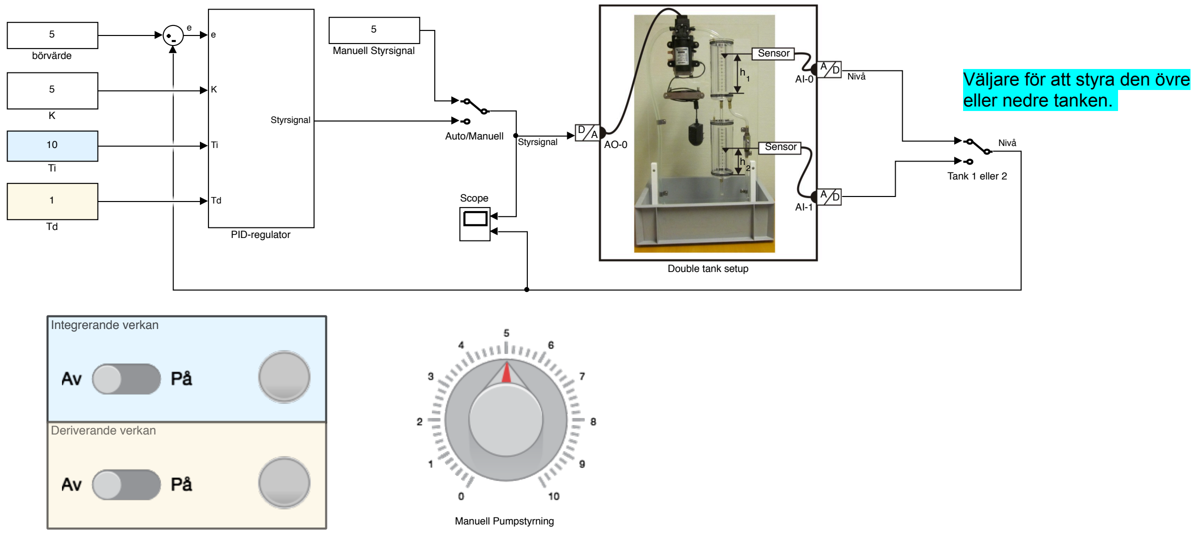


Figure 8: Simulink model.